Use-Case Flow: Redeem Digital Gift Card

Problem Statement #34 — Digital Gift Card & Loyalty Point Exchange

Use Case

UC-04: Redeem Digital Gift Card (includes UC-05: Lock Voucher)

Actors

Primary Actor: Account Holder — initiates the redemption and receives the gift card voucher.

Secondary Actor: Merchant Partner — validates and honors the voucher at the point of sale.

Preconditions

1. The Account Holder is logged into the wallet platform with at least one linked Merchant Partner loyalty account.
2. The Account Holder has a sufficient converted exchange-credit balance to cover the gift card value requested.

Postconditions

Success: A single-use, checksum-verified voucher is generated and locked, the Account Holder's exchange-credit balance is debited, and the transaction is recorded in the ledger with zero balance drift.

Failure: No voucher is generated, no balance is debited, and the failed attempt is logged.

Main Success Scenario

1. Account Holder selects "Redeem Gift Card" and chooses a Merchant Partner and denomination.
2. System checks the Account Holder's converted exchange-credit balance against the requested amount.
3. System validates the request and reserves the corresponding credits.
4. System generates a single-use digital voucher code with a cryptographic checksum.
5. System immediately locks the voucher so it cannot be redeemed a second time — «include» UC-05: Lock Voucher.
6. System debits the Account Holder's exchange-credit balance and records the transaction in the ledger.
7. System displays the voucher code and sends a confirmation notification to the Account Holder.
8. At the point of sale, the Merchant Partner submits the voucher code for validation.
9. System confirms the voucher is valid, unused, and unexpired, and returns a "Valid" status within 250 ms.
10. Merchant Partner honors the redemption; system marks the voucher as redeemed.
11. Use case ends successfully.

Alternate Flow

2a. Insufficient Balance

2a1. System displays an "Insufficient Balance" message and shows the current exchange-credit balance.

2a2. Account Holder is prompted to convert additional loyalty points (UC-03) or reduce the requested denomination.

2a3. If the Account Holder does not take either action, the redemption request is cancelled and no credits are reserved.

Alternate Flow (Merchant-Side)

9a. Voucher Fails Validation

9a1. System returns an "Invalid" status if the voucher is expired, already redeemed, or fails checksum verification.

9a2. Merchant Partner declines the redemption and informs the Account Holder.

9a3. System logs the failed validation attempt, including merchant ID and timestamp, for fraud review.